School of Engineering and Applied Sciences, Ahmedabad University

CSE 400: Fundamentals of Probability in Computing

Lecture 6: Discrete RVs, Expectation and Problem Solving

1 Previous Lecture Recap: Random Variables (RVs)

1.1 Motivation and Concept of a Random Variable

A **random variable** is a variable whose value depends on the outcome of a random experiment.

From the lecture:

- The **distribution of a random variable** can be visualized as a **bar diagram**.
- The **x-axis** represents the values that the random variable can take.
- The **height of the bar** at value a_i represents

$$\Pr[X = a_i]$$

- Each probability is computed by finding the probability of the **corresponding event in the sample space**.

1.2 Random Variable Example

Example 1: Tossing 3 Fair Coins

Let:

- The experiment consist of tossing **3 fair coins**
- Y = number of heads observed

Possible values of Y:

$$Y \in \{0, 1, 2, 3\}$$

Step-by-step computation of probabilities:

- $\Pr(Y = 0) = \Pr(t, t, t) = \frac{1}{8}$
- $\Pr(Y = 1) = \Pr(t, t, h), \Pr(t, h, t), \Pr(h, t, t) = \frac{3}{8}$
- $\Pr(Y = 2) = \Pr(h, h, t), \Pr(h, t, h), \Pr(t, h, h) = \frac{3}{8}$
- $\Pr(Y = 3) = \Pr(h, h, h) = \frac{1}{8}$

Verification:

$$\sum_{i=0}^{3} \Pr(Y = i) = \frac{1}{8} + \frac{3}{8} + \frac{3}{8} + \frac{1}{8} = 1$$

2 Probability Mass Function (PMF)

2.1 Concept

A random variable that can take **at most a countable number of values** is called a **discrete random variable**.

Let:

- X be a discrete random variable
- Range:

$$R_X = \{x_1, x_2, x_3, \dots\}$$

The function:

$$P_X(x_k) = \Pr(X = x_k), \quad k = 1, 2, 3, \dots$$

is called the **Probability Mass Function (PMF)** of X.

Property:

$$\sum_{k=1}^{\infty} P_X(x_k) = 1$$

2.2 PMF Example (Given in Lecture)

The PMF of a random variable X is given by:

$$p(i) = c\frac{\lambda^i}{i!}, \quad i = 0, 1, 2, \dots$$

where $\lambda > 0$.

Step 1: Use normalization condition

$$\sum_{i=0}^{\infty} p(i) = 1$$

$$c\sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = 1$$

Using:

$$e^{\lambda} = \sum_{i=0}^{\infty} \frac{\lambda^i}{i!}$$

So:

$$ce^{\lambda} = 1 \Rightarrow c = e^{-\lambda}$$

Step 2: Compute $\Pr[X = 0]$

$$\Pr[X = 0] = e^{-\lambda}\frac{\lambda^0}{0!} = e^{-\lambda}$$

Step 3: Compute $\Pr[X > 2]$

$$\Pr[X > 2] = 1 - \Pr[X \leq 2]$$

$$= 1 - (\Pr[X = 0] + \Pr[X = 1] + \Pr[X = 2])$$

$$= 1 - \left(e^{-\lambda} + \lambda e^{-\lambda} + \frac{\lambda^2}{2} e^{-\lambda}\right)$$

3 Independent Events

3.1 Definition

Two events A and B are **independent** if:

$$\Pr(A \cap B) = \Pr(A)\Pr(B)$$

4 Bayes' Theorem (Recap)

4.1 Bayes' Formula

Using:

$$\Pr(A \cap B) = \Pr(B|A)\Pr(A)$$

We obtain:

$$\Pr(B_i|A) = \frac{\Pr(A|B_i)\Pr(B_i)}{\sum_{j=1}^{n} \Pr(A|B_j)\Pr(B_j)}$$

This is known as **Bayes' Formula (Proposition 3.1)**.
Definitions:

- $\Pr(B_i)$: a priori probability
- $\Pr(B_i|A)$: posterior probability

4.2 Bayes' Theorem Example 1: Auditorium with 30 Rows

Given:

- 30 rows
- Row 1 has 11 seats
- Row 2 has 12 seats
- ...
- Row 30 has 40 seats

Procedure:

1. A row is selected uniformly at random
2. A seat is selected uniformly within the chosen row

Compute:

- Pr(Seat 15 | Row 20)
- Pr(Row 20 | Seat 15)

$$\Pr(S_{15}|R_{20}) = \frac{1}{30}$$

$$\Pr(S_{15}) = \sum_{k=5}^{30} \Pr(S_{15}|R_k)\Pr(R_k)$$

$$= \sum_{k=5}^{30} \frac{1}{k+10} \cdot \frac{1}{30}$$

$$\Pr(R_{20}|S_{15}) = \frac{\Pr(S_{15}|R_{20})\Pr(R_{20})}{\Pr(S_{15})}$$

5 Types of Discrete Random Variables

- **Bernoulli Random Variable**: Single trial, two outcomes
- **Binomial Random Variable**: Fixed number of trials, independent trials, same probability of success
- **Geometric Random Variable**: Counts trials until first success
- **Poisson Random Variable**: Models count of events in fixed interval

6 Expectation of Random Variables

6.1 Definition

$$\mathbb{E}[X] = \sum_x x \Pr(X = x)$$

6.2 Expectation of a Function of a Random Variable

$$\mathbb{E}[g(X)] = \sum_x g(x)\Pr(X = x)$$

6.3 Linearity of Expectation

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$$

6.4 Moments and Central Moments

- Variance:

$$\mathrm{Var}(X) = \mathbb{E}[(X - \mu)^2]$$

- Skewness: third central moment
- Kurtosis: fourth central moment

7 Cumulative Distribution Function (CDF)

7.1 Definition

$$F_X(x) = \Pr(X \leq x)$$

7.2 Properties

- Non-decreasing
- $\lim_{x \to -\infty} F_X(x) = 0$
- $\lim_{x \to \infty} F_X(x) = 1$

8 Probability Density Function (PDF)

8.1 Definition

$$f_X(x) = \frac{d}{dx} F_X(x)$$

8.2 Property

$$\int_{-\infty}^{\infty} f_X(x)\, dx = 1$$

End of Lecture 6 Scribe — Strictly Aligned with Course Slides